

```

1)
#!/bin/bash
echo "Enter a number:"
read num
if [  $((num \% 2)) -eq 0$  ]
then
    echo "$num is Even."
else
    echo "$num is Odd."
Fi

```

```

dragonfly@dragonfly-VMware-Virtual-Platform:~$ bash odd_even.sh
Enter a number:
2
2 is Even.

```

```

2)
#!/bin/bash
echo "Enter a number:"
read num
fact=1
for ((i=1; i<=num; i++))
do
    fact=$((fact * i))
done
echo "Factorial of $num is $fact"

```

```

dragonfly@dragonfly-VMware-Virtual-Platform:~$ bash factorial.sh
Enter a number:
5
Factorial of 5 is 120

```

```

3)
#!/bin/bash
echo "Enter number of terms:"
read n
a=0
b=1
echo "Fibonacci Series:"
for ((i=0; i<n; i++))
do
    echo -n "$a "
    temp=$((a+b))
    a=$b
    b=$temp
done
#!/bin/bash
read -p "Enter the number of terms: " n

```

```

a=0
b=1
count=0
echo "Fibonacci Series up to $n terms:"
while [ $count -lt $n ]
do
    echo -n "$a "
    next=$((a + b))
    a=$b
    b=$next
    count=$((count + 1))
done
echo ""echo
Output will be same

```

```

dragonfly@dragonfly-VMware-Virtual-Platform:~$ bash Fibonacci.sh
Enter number of terms:
10
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34

```

```

4)
#!/bin/bash
echo "Enter a number:"
read num
flag=0
for ((i=2; i<num; i++))
do
    if [ $((num%i)) -eq 0 ]
    then
        flag=1
        break
    fi
done
if [ $num -le 1 ]
then
    echo "$num is not Prime."
elif [ $flag -eq 0 ]
then
    echo "$num is Prime."
else
    echo "$num is not Prime."
Fi

```

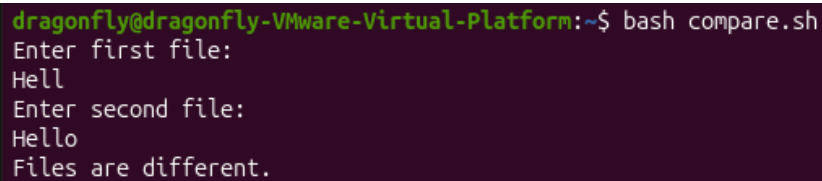
```

dragonfly@dragonfly-VMware-Virtual-Platform:~$ bash prime.sh
Enter a number:
3
3 is Prime.

```

5)

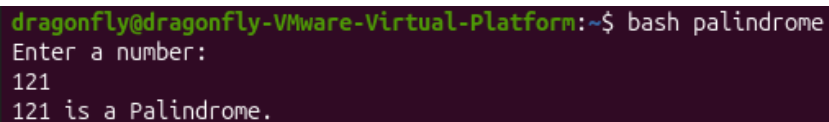
```
#!/bin/bash
echo "Enter first file:"
read file1
echo "Enter second file:"
read file2
if cmp -s "$file1" "$file2"
then
    echo "Files are identical."
    rm "$file2"
    echo "$file2 deleted."
else
    echo "Files are different."
Fi
```



```
dragonfly@dragonfly-VMware-Virtual-Platform:~$ bash compare.sh
Enter first file:
Hell
Enter second file:
Hello
Files are different.
```

6)

```
#!/bin/bash
echo "Enter a number:"
read num
original=$num
reverse=0
while [ $num -gt 0 ]
do
    digit=$((num % 10))
    reverse=$((reverse * 10 + digit))
    num=$((num / 10))
done
if [ $original -eq $reverse ]
then
    echo "$original is a Palindrome."
else
    echo "$original is not a Palindrome."
Fi
```



```
dragonfly@dragonfly-VMware-Virtual-Platform:~$ bash palindrome
Enter a number:
121
121 is a Palindrome.
```

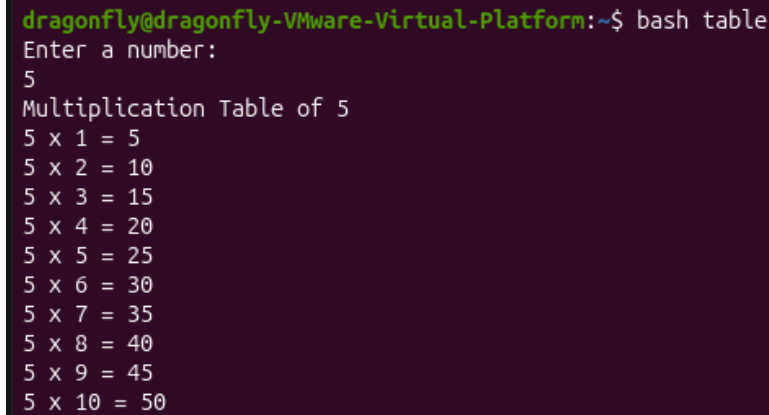
7)

```
#!/bin/bash
echo "Enter a number:"
read num
```

```

echo "Multiplication Table of $num"
for ((i=1; i<=10; i++))
do
    echo "$num x $i = $((num*i))"
done

```



```

dragonfly@dragonfly-VMware-Virtual-Platform:~$ bash table
Enter a number:
5
Multiplication Table of 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

```

```

8)
#!/bin/bash
if [ $# -lt 1 ]; then
    echo "Usage: $0 <directory>"
    exit 1
fi
DIR="$1"
if [ ! -d "$DIR" ]; then
    echo "Error: '$DIR' is not a valid directory"
    exit 1
fi
mapfile -t FILES < <(find "$DIR" -type f -name "*.log" -mtime -7 2>/dev/null)
FILE_COUNT=${#FILES[@]}
if [ "$FILE_COUNT" -eq 0 ]; then
    echo "Directory scanned: $DIR"
    echo "Log files found: 0"
    echo "Total lines: 0"
    echo "No .log files modified in the last 7 days were found."
    exit 0
fi
TOTAL_LINES=0
declare -A LINE_COUNTS
for f in "${FILES[@]}"; do
    lines=$(wc -l < "$f" 2>/dev/null)
    lines=${lines:-0}
    LINE_COUNTS["$f"]=$lines
    TOTAL_LINES=$((TOTAL_LINES + lines))
done

```

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done
echo "Directory scanned: $DIR"
echo "Log files found: $FILE_COUNT"
echo "Total lines: $TOTAL_LINES"
echo ""
echo "Files (sorted by name):"
for f in $(printf '%s\n' "${FILES[@]}" | sort); do
    printf "%-60s %8d lines\n" "$f" "${LINE_COUNTS[$f]}"
done

```

```

hitten@Hitten:~$ bash log_finder.sh /var/log
Directory scanned: /var/log
Log files found: 4
Total lines: 1953

Files (sorted by name):
/var/log/auth.log          97 lines
/var/log/kern.log         1856 lines
/var/log/landscape/landscape-config.log    0 lines
/var/log/unattended-upgrades/unattended-upgrades-shutdown.log 0 lines

```

```

9)#!/usr/bin/env bash
#
# sys_report.sh - Collects system health information and saves it to a
# timestamped report file. Simulates a periodic system snapshot that can
# help spot anomalies over time.
#
# Usage: ./sys_report.sh
#
set -uo pipefail
# -----
# Setup
# -----
SECONDS=0 # bash builtin, auto-increments every second since script start
TIMESTAMP="$(date +%Y%m%d_%H%M%S)"
OUTFILE="sys_report_${TIMESTAMP}.txt"
# -----
# Helpers
# -----
section_header() {
    # $1 = section title
    echo ""
    echo
    "=====
    echo " $1"
    echo
    "=====
}

```

```

# -----
# Section functions
# -----

report_header() {
    section_header "SYSTEM HEALTH REPORT"
    echo "Generated at : $(date '+%Y-%m-%d %H:%M:%S %Z')"
```

echo "Hostname : \$(hostname)"

echo "Current user : \$(whoami)"

```

}

report_uptime() {
    section_header "SYSTEM UPTIME"
    if command -v uptime >/dev/null 2>&1; then
        uptime
    else
        echo "uptime command not available"
    fi
}

report_disk_usage() {
    section_header "DISK USAGE (Home Directory Mount)"
    if command -v df >/dev/null 2>&1; then
        # Print header line, then only the row for the mount containing $HOME
        df -h | head -1
        df -h "$HOME" | tail -1
    else
        echo "df command not available"
    fi
}

report_memory_usage() {
    section_header "MEMORY USAGE"
    if command -v free >/dev/null 2>&1; then
        free -h
    else
        echo "free command not available"
    fi
}

report_processes() {
    section_header "RUNNING PROCESSES"
    if command -v ps >/dev/null 2>&1; then
        local total
        total=$(ps aux | tail -n +2 | wc -l)
        echo "Total running processes: ${total}"
        echo ""
        echo "Top 5 by CPU usage:"
    fi
}

```

```

        ps aux --sort=-%cpu | head -6
    else
        echo "ps command not available"
    fi
}
report_network() {
    section_header "ACTIVE NETWORK CONNECTIONS (Listening Ports)"
    if command -v ss >/dev/null 2>&1; then
        ss -tulpn 2>/dev/null
    elif command -v netstat >/dev/null 2>&1; then
        netstat -tulpn 2>/dev/null
    else
        echo "Neither ss nor netstat is available"
    fi
}
report_logins() {
    section_header "RECENT LOGINS (Last 5)"
    if command -v last >/dev/null 2>&1; then
        last | head -5
    else
        echo "last command not available"
    fi
}
report_footer() {
    section_header "END OF REPORT"
    echo "Total generation time: ${SECONDS} second(s)"
}
# -----
# Main
# -----
main() {
    exec 3>&1
    exec > "$OUTFILE"
    report_header
    report_uptime
    report_disk_usage
    report_memory_usage
    report_processes
    report_network
    report_logins
    report_footer
    exec 1>&3 3>&-
    echo "Report generated: $OUTFILE"
}

```

main "\$@"

```
dragonfly@dragonfly-Vmware-Virtual-Platform:~$ bash sys.sh
Report generated: sys_report_20260701_121823.txt
dragonfly@dragonfly-Vmware-Virtual-Platform:~$ cat sys_report_20260701_121823.txt

=====
SYSTEM HEALTH REPORT
=====
Generated at : 2026-07-01 12:18:23 IST
Hostname      : dragonfly-Vmware-Virtual-Platform
Current user  : dragonfly

=====
SYSTEM UPTIME
=====
12:18:23 up 4 min,  1 user,  load average: 0.88, 0.73, 0.33

=====
DISK USAGE (Home Directory Mount)
=====
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2        28G   12G   6.8G   64% /

=====
MEMORY USAGE
=====
              total        used        free      shared  buff/cache   available
Mem:          3.3Gi        1.9Gi        253Mi        73Mi        1.3Gi        1.4Gi
Swap:         3.3Gi        139Mi        3.2Gi

=====
RUNNING PROCESSES
=====
Total running processes: 369

Top 5 by CPU usage:
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
dragon+  3164 16.9 13.6 12153252 467184 ?        SL   12:16   0:15 /snap/firefox/8504/usr/lib/firefox/firefox
dragon+  3904  9.0 12.1 7394172 417668 ?        SL   12:16   0:07 /snap/firefox/8504/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle 0:46599 -prefMapHandle 1:287079 -jsInitHandle 2:15
0120 -parentBuildID 20260615155924 -sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {160c8de9-b90f-4096-9dbc-71f4c17ff71e} -parentPid 3164 -crashReporter 0 -crashHelperPid 3237 -crash
Helper 7 -greomni /snap/firefox/8504/usr/lib/firefox/omni.ja -appomni /snap/firefox/8504/usr/lib/firefox/browser/omni.ja -appDir /snap/firefox/8504/usr/lib/firefox/browser 7 tab
dragon+  2177  4.1  7.1 3816372 246328 ?        Ssl  12:14   0:10 /usr/bin/gnome-shell --mode=ubuntu
dragon+  3342  2.1  6.3 2779868 218360 ?        SL   12:16   0:01 /snap/firefox/8504/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle 0:35080 -prefMapHandle 1:287079 -jsInitHandle 2:15
0120 -parentBuildID 20260615155924 -sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {f7f55495-19d9-4d75-9a25-79f8a385fa83} -parentPid 3164 -crashReporter 0 -crashHelperPid 3237 -crash
Helper 7 -greomni /snap/firefox/8504/usr/lib/firefox/omni.ja -appomni /snap/firefox/8504/usr/lib/firefox/browser/omni.ja -appDir /snap/firefox/8504/usr/lib/firefox/browser 3 tab
dragon+  3091  1.5  6.3 1613668 218388 ?        SL   12:14   0:03 /usr/bin/ptxix --gapplication-service

=====
ACTIVE NETWORK CONNECTIONS (Listening Ports)
=====
NetId State  Recv-Q Send-Q Local Address:Port Peer Address:PortProcess
udp UNCONN 0 0 127.0.0.54:53 0.0.0.0:*
udp UNCONN 0 0 127.0.0.53%lo:53 0.0.0.0:*
udp UNCONN 0 0 :::0.0.1:323 0.0.0.0:*

/dev/sda2        28G   12G   6.8G   64% /

=====
MEMORY USAGE
=====
              total        used        free      shared  buff/cache   available
Mem:          3.3Gi        1.9Gi        253Mi        73Mi        1.3Gi        1.4Gi
Swap:         3.3Gi        139Mi        3.2Gi

=====
RUNNING PROCESSES
=====
Total running processes: 369

Top 5 by CPU usage:
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
dragon+  3164 16.9 13.6 12153252 467184 ?        SL   12:16   0:15 /snap/firefox/8504/usr/lib/firefox/firefox
dragon+  3904  9.0 12.1 7394172 417668 ?        SL   12:16   0:07 /snap/firefox/8504/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle 0:46599 -prefMapHandle 1:287079 -jsInitHandle 2:15
0120 -parentBuildID 20260615155924 -sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {160c8de9-b90f-4096-9dbc-71f4c17ff71e} -parentPid 3164 -crashReporter 0 -crashHelperPid 3237 -crash
Helper 7 -greomni /snap/firefox/8504/usr/lib/firefox/omni.ja -appomni /snap/firefox/8504/usr/lib/firefox/browser/omni.ja -appDir /snap/firefox/8504/usr/lib/firefox/browser 7 tab
dragon+  2177  4.1  7.1 3816372 246328 ?        Ssl  12:14   0:10 /usr/bin/gnome-shell --mode=ubuntu
dragon+  3342  2.1  6.3 2779868 218360 ?        SL   12:16   0:01 /snap/firefox/8504/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle 0:35080 -prefMapHandle 1:287079 -jsInitHandle 2:15
0120 -parentBuildID 20260615155924 -sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {f7f55495-19d9-4d75-9a25-79f8a385fa83} -parentPid 3164 -crashReporter 0 -crashHelperPid 3237 -crash
Helper 7 -greomni /snap/firefox/8504/usr/lib/firefox/omni.ja -appomni /snap/firefox/8504/usr/lib/firefox/browser/omni.ja -appDir /snap/firefox/8504/usr/lib/firefox/browser 3 tab
dragon+  3091  1.5  6.3 1613668 218388 ?        SL   12:14   0:03 /usr/bin/ptxix --gapplication-service

=====
ACTIVE NETWORK CONNECTIONS (Listening Ports)
=====
NetId State  Recv-Q Send-Q Local Address:Port Peer Address:PortProcess
udp UNCONN 0 0 127.0.0.54:53 0.0.0.0:*
udp UNCONN 0 0 127.0.0.53%lo:53 0.0.0.0:*
udp UNCONN 0 0 127.0.0.1:323 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:50287 0.0.0.0:* users:(("firefox",pid=3164,fd=61))
udp UNCONN 0 0 0.0.0.0:5353 0.0.0.0:*
udp UNCONN 0 0 [::]:323 [::]:*
udp UNCONN 0 0 [::]:5353 [::]:*
tcp LISTEN 0 4096 127.0.0.1:631 0.0.0.0:*
tcp LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:*
tcp LISTEN 0 4096 127.0.0.54:53 0.0.0.0:*
tcp LISTEN 0 4096 [::]:631 [::]:*

=====
RECENT LOGINS (Last 5)
=====
last command not available

=====
END OF REPORT
=====
Total generation time: 0 second(s)
dragonfly@dragonfly-Vmware-Virtual-Platform:~$
```

10)
#!/bin/bash
LENGTH=\${1:-16}


```

COUNT=${2:-1}
if ! [[ "$LENGTH" =~ ^[0-9]+$ ]] || ! [[ "$COUNT" =~ ^[0-9]+$ ]]; then
    exit 1
fi
if (( LENGTH < 8 || LENGTH > 128 )) || (( COUNT < 1 || COUNT > 100 )); then
    exit 1
fi
CHAR_POOL='A-Za-z0-9!@#%&^*\(\)_-=+~\[\]\{\}\|;,:.<>\?'
LOG_FILE="passwords.log"
for ((i = 1; i <= COUNT; i++)); do
    PASSWORD=$(LC_ALL=C tr -dc "$CHAR_POOL" < /dev/urandom | head -c "$LENGTH")
    echo "$PASSWORD"
    echo "[$(date '+%Y-%m-%d %H:%M:%S')] $PASSWORD" >> "$LOG_FILE"
done

```

```

dragonfly@dragonfly-VMware-Virtual-Platform:~$ bash gen_pass.sh
OPD2m|gfJJvy]1*5

```